

Dongjun (Joyce) Xie

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SUMMARY

AI-focused full-stack engineer building production-grade multi-agent systems, agent runtimes, and LLM applications. Skilled in context engineering, tool orchestration, memory, evaluation, and AWS/GCP deployment.

EDUCATION

Northeastern University, San Jose, CA	September 2022 - August 2025
Master of Science in Computer Science (GPA: 4.0/4.0)	
University of California, Irvine, Irvine, CA	September 2015 - March 2019
Bachelor of Arts in Quantitative Economics (GPA: 3.5/4.0)	

TECHNICAL SKILLS

AI & Agent Development: Multi-agent orchestration, tool use, context engineering, agent memory, evaluation, guardrails, structured outputs, MCP, prompt caching.

Programming Languages: TypeScript, Python, Rust, JavaScript, SQL, HTML/CSS.

Frameworks: Next.js, React, FastAPI, Node.js, Hono, Vercel AI SDK, Playwright.

Cloud & Data: AWS (CDK, Lambda, SQS, DynamoDB, CloudWatch), GCP, Docker, PostgreSQL, SQLite, Redis.

PROFESSIONAL EXPERIENCE

SiriusMindshare LLC, San Jose, CA	October 2025 - Present
Machine Learning Engineer <i>Next.js, TypeScript, AWS, Vercel AI SDK</i>	

- Architected and shipped a production-grade multi-agent platform with **5 specialized agents** spanning marketing, design, bookkeeping, support, and sales, each with distinct personas, tools, and memory scope.
- Engineered a **cache-aware prompt assembly pipeline** that reused ~4K static tokens per request to reduce inference latency and per-call LLM cost.
- Built a 4-stage **context compression and recovery system** triggered at 80% token capacity, preserving critical state across long-running agent sessions.
- Extended the same agent-evaluation and multimodal reasoning patterns into an **LVM retail attention-quality framework**; co-authored a paper accepted for publication and presentation at IEEE CCET 2026 (forthcoming).

1 Thing Against Racism Organization , Monroe, NJ	January 2024 - April 2025
Full-Stack Software Developer Intern (Remote) <i>React, React Native, Flask, Firebase, GCP</i>	

- Delivered cross-platform clients and cloud APIs with real-time data, OAuth, encryption, caching, notifications, and **AI-assisted content moderation**.

PROJECTS

AgentFlow TypeScript, Bun, React, Hono, WebSocket	2026
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- Built a graph-based **multi-agent orchestration platform** with typed steps, roles, gates, routers, parallel scheduling, human approvals, and real-time execution events.
- Implemented a contract-driven mission engine with **tool governance, token budgets, crash recovery, worktree isolation**, checkpoints, and evidence-backed delivery.

Agent Bridge Rust, Python, TypeScript, Docker, AWS	2026
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- Created an agent-first runtime that provisions **disposable sandboxed compute** across local Docker and AWS through a five-verb CLI, MCP surface, and Python/TypeScript SDKs.
- Added secrets injection, tiered artifacts, retries, timeouts, **cost accounting and bounded audit logs** for observable, framework-agnostic execution.

Kaggle Team TypeScript, Bun, Zod, Kaggle API	2026
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- Orchestrated Researcher, PM, and Engineer agents through **score-informed Kaggle loops** with typed tools, parallel strategies, context compaction, cost tracking, and CV-to-LB feedback.

Additional open source: Agent Hook (bounded event-to-agent triggers) | jsfetch (Playwright CLI/MCP for JS-rendered pages)